

Customer Shopping Behavior Analysis

Uncovering spending patterns, segments, and product preferences from
3,900 transactions



Dataset at a Glance

3,900

Transactions

18

Columns

37

Missing Values

Review Rating only

\$59.76

Avg Purchase

Features: demographics, purchase details, shopping behavior, subscription status

Data Preparation in Python

01

Load & Explore

pandas, df.info(), .describe()

02

Clean

Imputed missing Review Ratings by category median

03

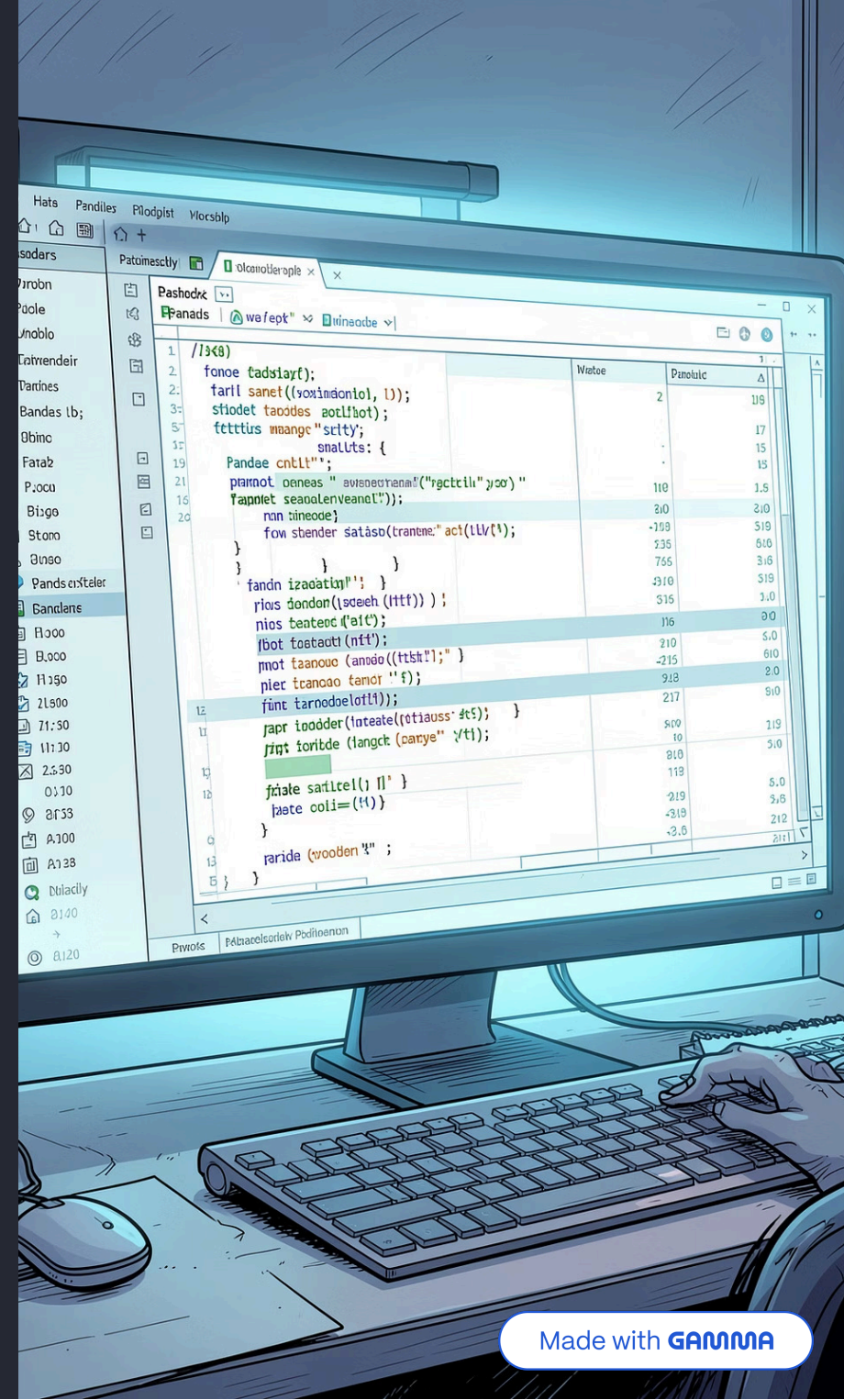
Engineer Features

age_group bins, purchase_frequency_days

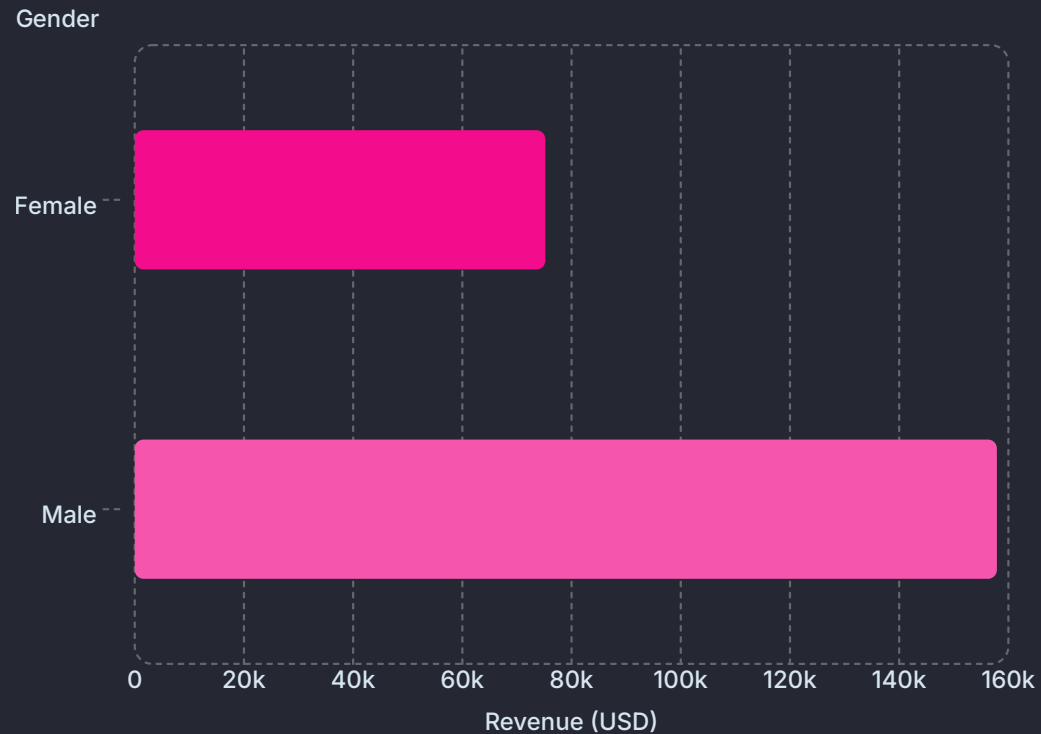
04

Load to PostgreSQL

Cleaned DataFrame ready for SQL



Revenue by Gender



Key Insight

Male customers generated **2.1x more revenue** than female customers — \$157,890 vs \$75,191

Top Products & Shipping Insights

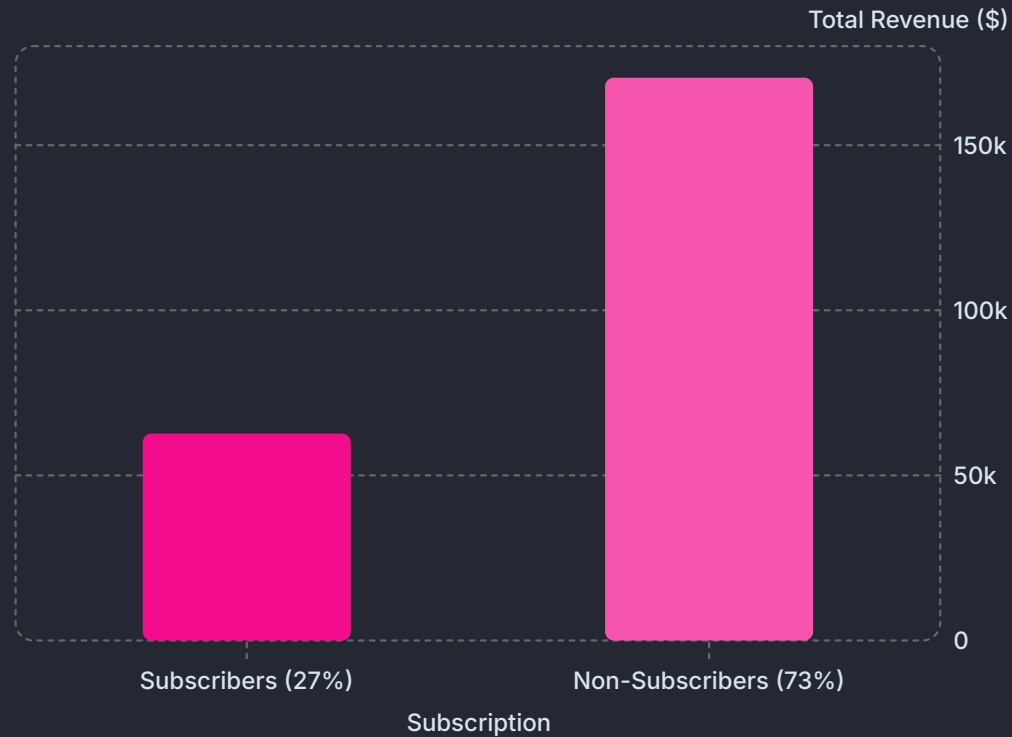
Top 5 Products by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Shipping Type Avg Spend



Subscribers vs. Non-Subscribers

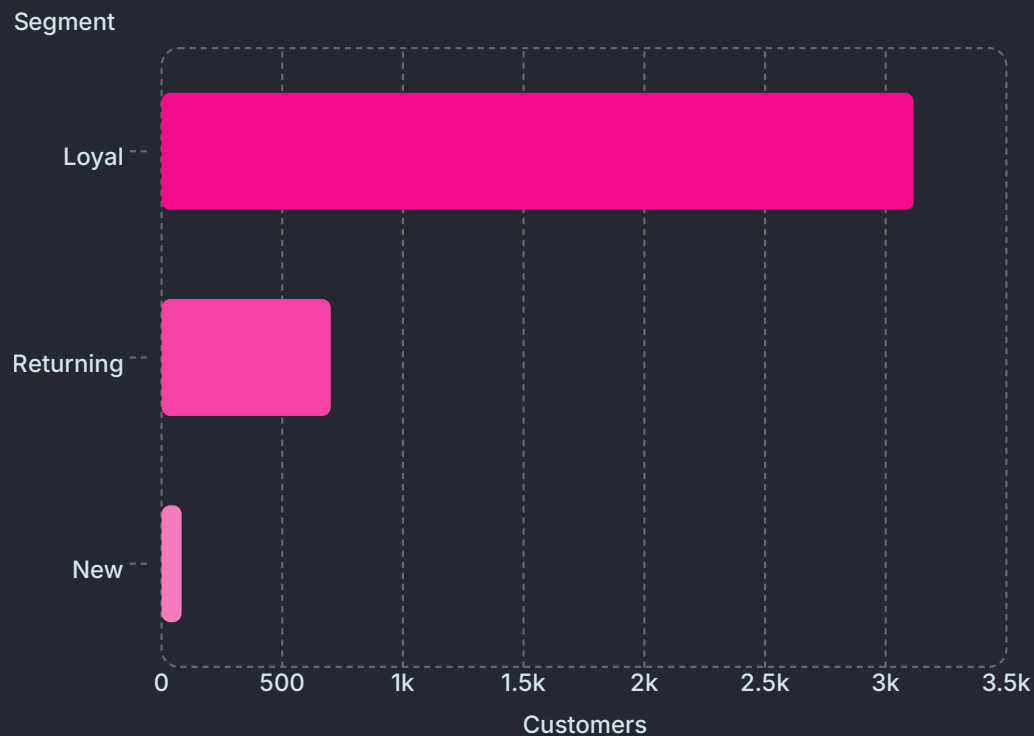


Subscriber Profile

- **1,053** subscribers — 27% of customers
- Avg spend: **\$59.49**
- Non-subscribers: **2,847** customers, avg \$59.87

📌 Similar avg spend — subscription growth is a volume opportunity

Customer Segmentation

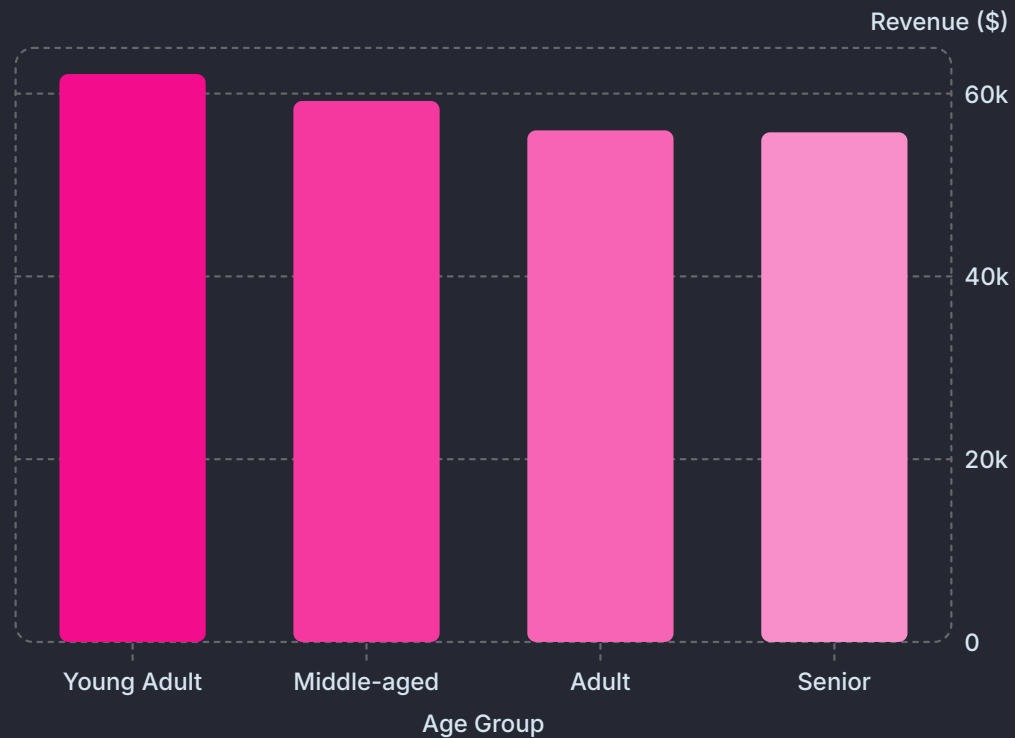


Segment Breakdown

- **Loyal:** 3,116 customers (80%)
- **Returning:** 701 customers
- **New:** 83 customers

958 repeat buyers (>5 purchases) hold subscriptions

Revenue by Age Group



Top Discount-Dependent Products

Hat

50% discount rate

Sneakers

49.66% discount rate

Coat

49.07% discount rate

Power BI Dashboard



3.9K

Total Customers

\$59.76

Avg Purchase

3.75

Avg Review Rating

Clothing

Top category: \$104K revenue

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → Loyal segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users